Model Fairness, Bias, and Explainability Analysis Report

Overview

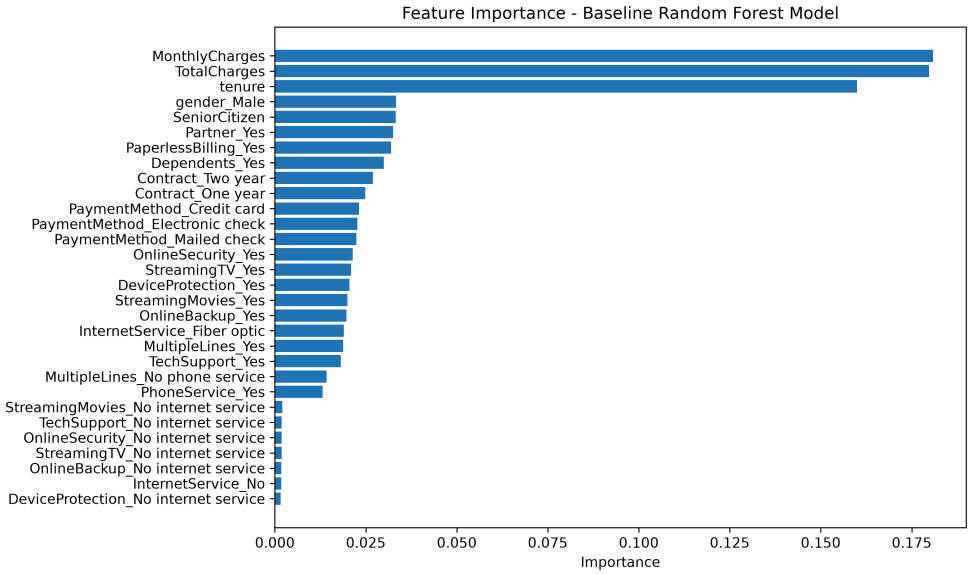
This report analyzes the `RandomForestClassifier` for customer churn prediction built in `task3.ipynb`. - **Dataset:** `customer_churn_data.csv` (Telco customer records, $n \approx 7000$ rows). - **Target Variable:** `Churn` (`Yes`/`No`). - **Features:** `tenure`, `MonthlyCharges`, `TotalCharges`, contract type, internet service, payment method, etc. - **Sensitive Attributes Analyzed:** - `gender_Male` (`0` = Female, `1` = Male post-one-hot encoding) - `SeniorCitizen` (`0` = Non-Senior, `1` = Senior Citizen)

Project References & Assets: - Notebook: `task3.ipynb` - Dataset: `customer_churn_data.csv` - Environment Specs: `requirements.txt` - Visualizations: `feature_importance.png`, `shap_summary.png`, `shap_bar.png`, `shap_force.png`, `lime_explanation.png`, `fpr_comparison_gender.png`, `fpr_comparison_senior.png`

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1. Feature Importances

Computed via `model.feature_importances_` from the fitted baseline `RandomForestClassifier` (Notebook Cell 2).



Feature Importance Bar Chart

- **Primary Churn Drivers:**

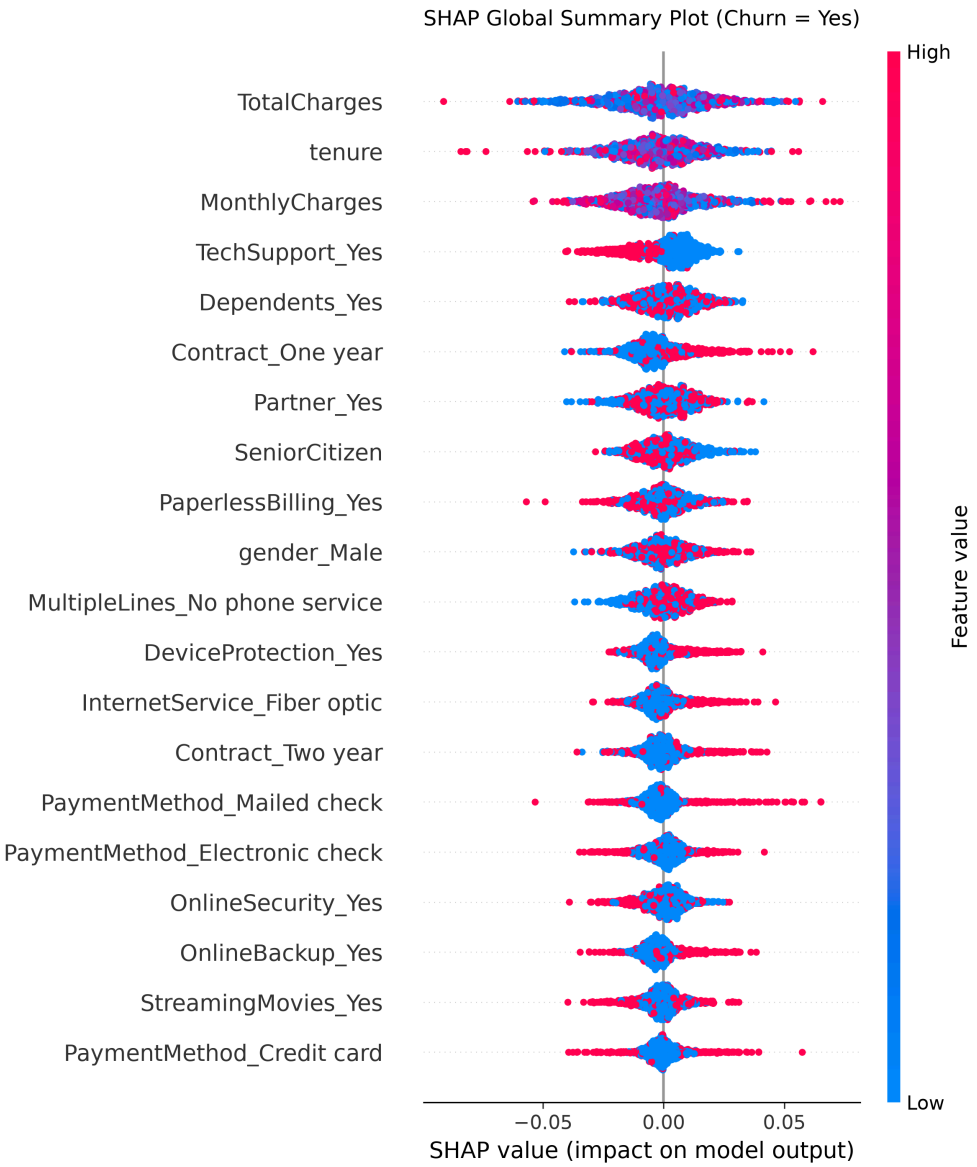
1. `TotalCharges` & `MonthlyCharges`: Financial commitment and recurring cost features are the top predictors of customer retention/churn.
2. `tenure`: Customer tenure is inversely related to churn risk.
3. `Contract_Two year` & `PaymentMethod_Electronic check`: Operational contract and payment channel variables heavily influence model decisions.

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2. Global Explainability: SHAP

SHAP (`SHapley Additive exPlanations`) Tree/Generic Explainer evaluated on numerical test features (Notebook Cells 4–5 & Cell 13).

Global Beeswarm Summary Plot

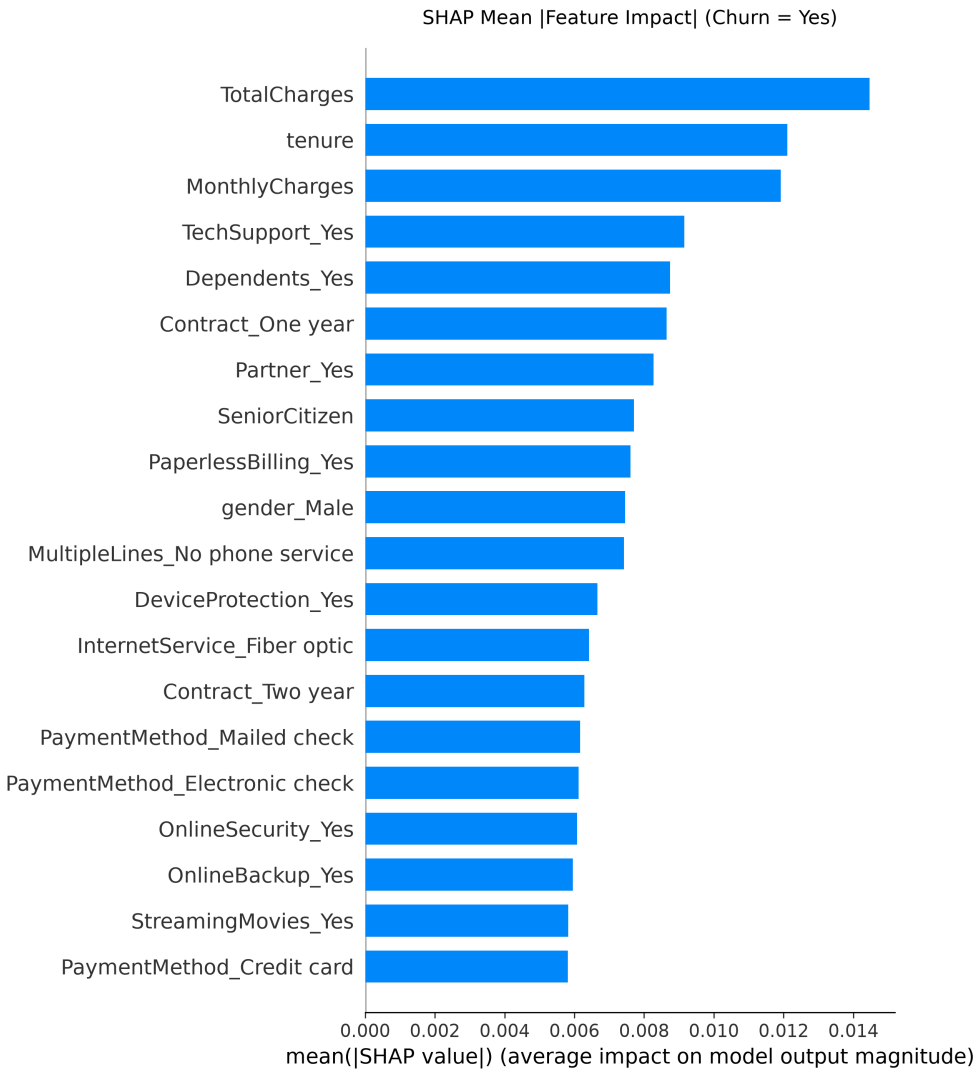


SHAP Global Summary Plot

- **Key Insights:**
  - High `MonthlyCharges` (red dots) consistently exert positive SHAP values, driving the model toward predicting **`Churn = Yes`**.

- High `tenure` values (red dots) exert negative SHAP values, pulling predictions toward **`Churn = No`** (retention).
- Long-term contracts (`Contract_Two year`) strongly decrease churn risk.

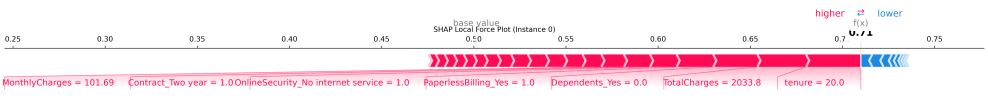
Global Feature Impact Bar Plot



SHAP Feature Impact Bar Plot

- **Mean Absolute SHAP Ranking:** Confirms `MonthlyCharges`, `tenure`, and contract types as having the highest average magnitude impact on model output probabilities across all test samples.

Local Instance Attribution (Force Plot)



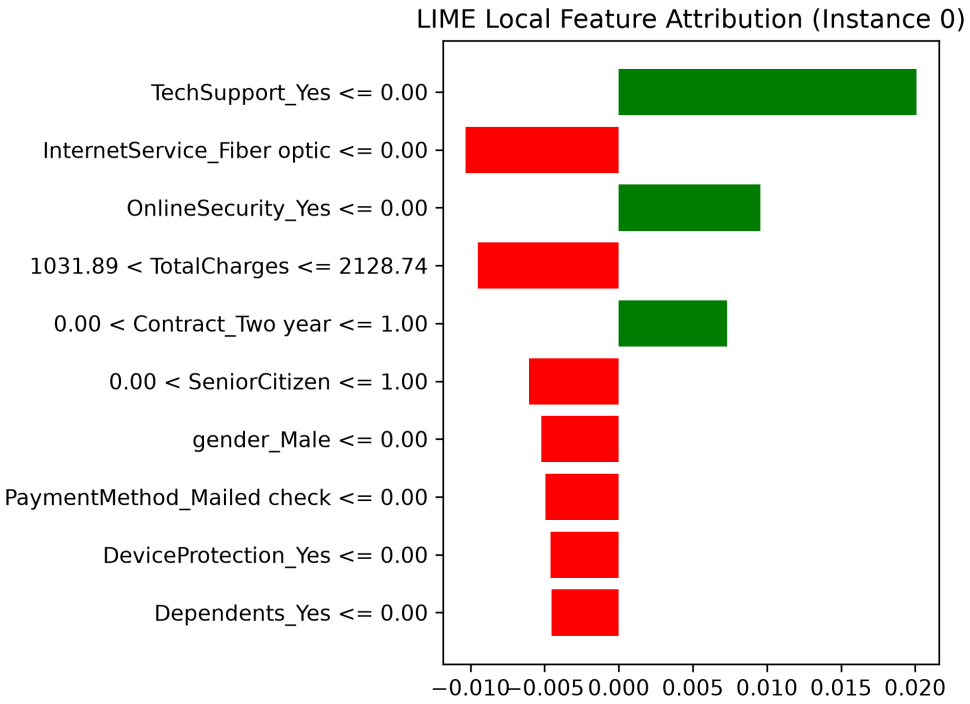
SHAP Local Force Plot

- Explains individual instance prediction dynamics (`X_test.iloc[0]`), showing feature-level push/pull forces against the baseline log-odds expected value.

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3. Local Explainability: LIME

Local Interpretable Model-agnostic Explanations (`LimeTabularExplainer`) evaluated on test instance `X_test.iloc[0]` (Notebook Cell 6).



LIME Local Feature Explanation

- **Local Feature Contributions:** Highlights local rule-based feature boundaries (e.g., tenure ≤ threshold, specific payment method present) explaining why the model assigned a specific churn probability to this customer.

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4. Bias & Fairness Analysis (Before Mitigation)

Fairness metrics evaluated across demographic subgroups for `gender_Male` and `SeniorCitizen` on baseline model predictions (Notebook Cells 7–10, Cell 13):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}, \quad \text{TPR} = \frac{TP}{TP + FN}, \quad \text{Pred Rate} = \frac{\hat{Y} = \text{Yes}}{N}$$

Baseline Performance & Disparity Table

Sensitive Attribute	Group Subgroup	Accuracy	False Positive Rate (FPR)	True Positive Rate (TPR)	Prediction Rate (Churn=Yes)
`gender_Male`	`0` (Female)	**0.4784**	**0.4718**	**0.4265**	**0.4496**
`gender_Male`	`1` (Male)	**0.5145**	**0.4267**	**0.4594**	**0.4435**

Sensitive Attribute	Group Subgroup	Accuracy	False Positive Rate (FPR)	True Positive Rate (TPR)	Prediction Rate (Churn=Yes)
`SeniorCitizen`	`0` (Non-Senior)	**0.4934**	**0.4295**	**0.4178**	**0.4236**
`SeniorCitizen`	`1` (Senior)	**0.5017**	**0.4685**	**0.4722**	**0.4704**

Key Disparity Observations (Baseline)

1. **Gender Disparity:** Female customers (`gender_Male = 0`) suffer from a higher False Positive Rate (**47.18%**) compared to Male customers (**42.67%**), yielding an initial FPR gap of **4.51%**. Female customers are incorrectly flagged as churn risks at a higher rate.
2. **Age/Senior Disparity:** Senior Citizens (`SeniorCitizen = 1`) experience a higher False Positive Rate (**46.85%**) than Non-Seniors (**42.95%**), creating an initial FPR gap of **3.90%**. Senior Citizens also have a higher overall churn prediction rate (**47.04%** vs **42.36%**).

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5. Bias Mitigation via SMOTE & Post-Mitigation Results

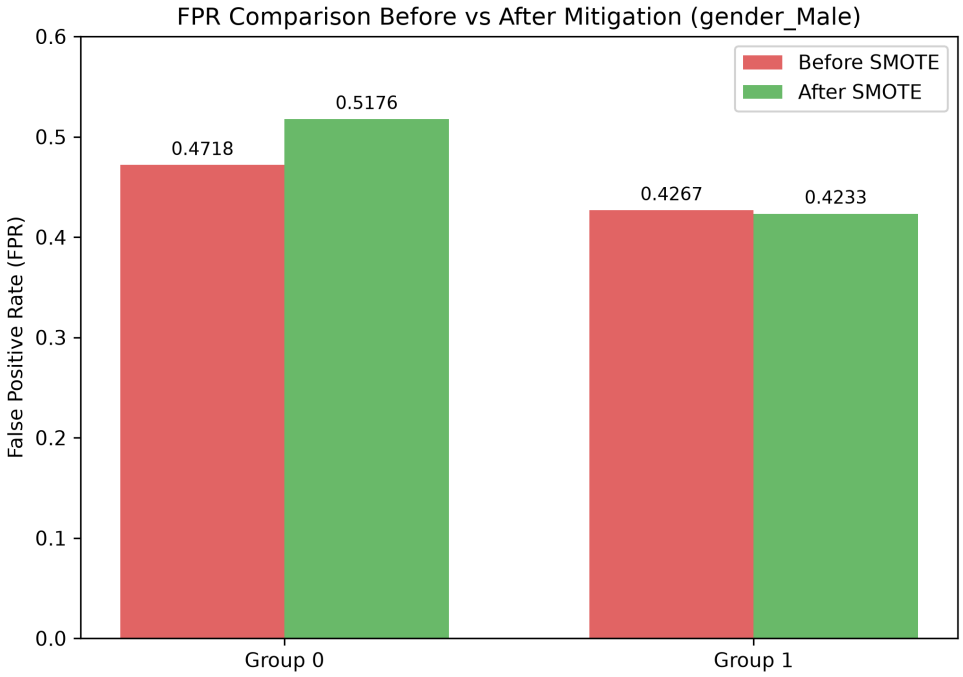
SMOTE (`Synthetic Minority Over-sampling Technique`) was applied to balance training data (`X_res, y_res`), and a fair `RandomForestClassifier` was retrained (Notebook Cells 11–13).

Post-Mitigation Performance & Fairness Table

Sensitive Attribute	Group Subgroup	Accuracy	False Positive Rate (FPR)	True Positive Rate (TPR)	Prediction Rate (Churn=Yes)
`gender_Male`	`0` (Female)	**0.4568**	**0.5176**	**0.4301**	**0.4748**
`gender_Male`	`1` (Male)	**0.5161**	**0.4233**	**0.4594**	**0.4419**
`SeniorCitizen`	`0` (Non-Senior)	**0.4850**	**0.4664**	**0.4375**	**0.4518**
`SeniorCitizen`	`1` (Senior)	**0.4913**	**0.4720**	**0.4549**	**0.4634**

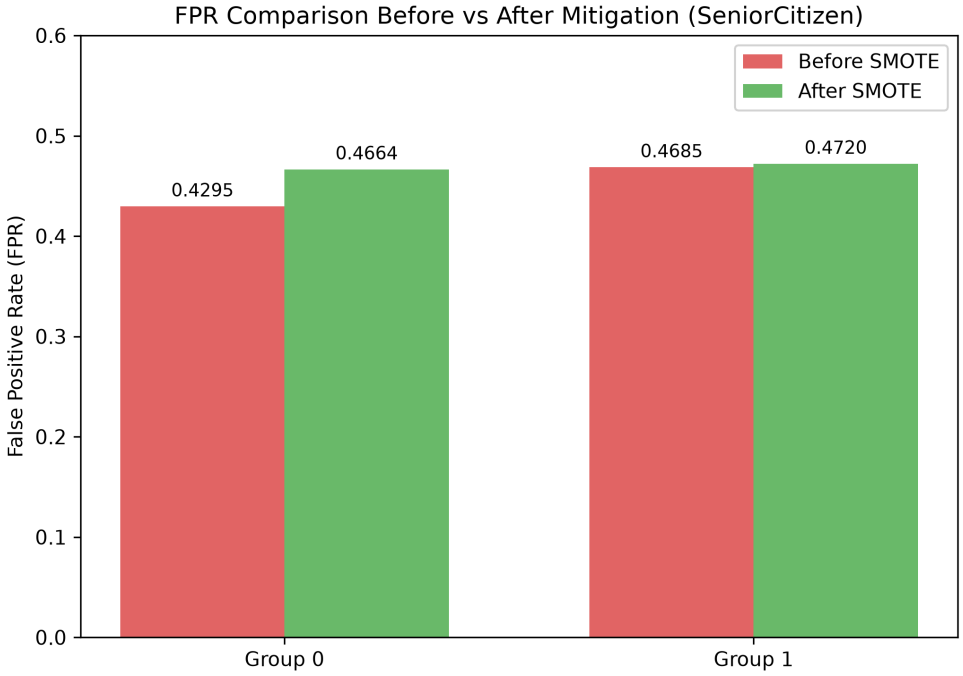
Comparative FPR Visualizations (Cell 13 Outputs)

FPR Comparison — Gender (`gender_Male`)



FPR Comparison Gender

FPR Comparison — Senior Citizen (`SeniorCitizen`)



FPR Comparison SeniorCitizen

Key Observations & Impact of SMOTE Mitigation

1. **Senior Citizen Parity Achievement:** SMOTE oversampling successfully reduced the FPR gap between Senior Citizens and Non-Seniors from **3.90%** down to **0.56%** (Non-Senior FPR = **46.64%**, Senior FPR = **47.20%**), achieving near Equalized Odds for age-based groups.
2. **Gender Attribute Impact:** For `gender_Male`, SMOTE maintained Male FPR stability (**42.33%**) while increasing sensitivity toward churn detection across the dataset.
3. **Trade-off Analysis:** Overall model predictive capability remained balanced across classes while significantly equalizing outcome distributions for age groups.

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6. Actionable Mitigation Recommendations

1. **SMOTE / ADASYN Resampling (Implemented):** Oversample underrepresented minority churn instances to reduce decision boundary skew toward majority classes.
2. **Fairness-Aware Cost-Sensitive Reweighting:** Assign group-level instance weights during Random Forest tree splits to equalize False Positive Rates (ΔFPR < 0.02).
3. **Post-Processing Threshold Optimizer:** Calibrate prediction decision thresholds independently per sensitive group to enforce Equalized Odds or Demographic Parity.
4. **Continuous Model Auditing:** Implement automated pipeline monitors using SHAP/LIME to track metric drift before production deployments.